

Rakesh R

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Summary — Full Stack Developer specializing in **React, Node.js, and AI-integrated systems**, with production experience in scalable, cloud-native automation systems. IEEE EPICS grant recipient with a strong track record of delivering scalable, end-to-end web solutions. Proficient in system design, RESTful API development, RAG pipelines, and cloud deployment. Passionate about engineering reliable, high-impact software for enterprise-scale environments.

Skills

Frontend	React.js, JavaScript (ES6+), Tailwind CSS, HTML5/CSS3
Backend	Node.js, Java, Python, REST API Design, JWT Authentication, JDBC
Databases	MongoDB, MySQL, Firebase, Vector Databases
Cloud & DevOps	AWS, Docker, Vercel, Netlify, GitHub, Postman
Generative AI	LLM APIs, LangChain, LlamaIndex, RAG Pipelines, Embeddings, OpenAI API
Practices	Agile/Scrum, System Design, RESTful Architecture, CI/CD, Code Review, TCP/IP

Projects

1. NidhiNanban – AI-Powered Personal Finance Coach

React.js, TypeScript, FastAPI, SQLite, Scikit-learn, SQLAlchemy, Telegram Bot API, Recharts

- Architected and deployed a full-stack AI-powered personal finance platform enabling users to monitor expenses, assess financial wellness through health scoring, plan savings goals, and visualize spending trends using interactive analytical dashboards.
- Developed an ML-based transaction intelligence engine utilizing TF-IDF vectorization and Logistic Regression for automated expense categorization; integrated Telegram-based conversational expense logging and spending trend forecasting to provide real-time financial insights and personalized coaching recommendations.

2. FluentPluz – Learning Management System

React.js, Node.js, MongoDB, Firebase

- Engineered a tailored LMS for a Japanese language institute serving 100+ active learners, enabling structured course delivery, module-level assessments, and real-time student progress tracking.
- Implemented role-based access control (Admin/Instructor/Student), reducing manual administrative workflows by ~40% and improving content delivery consistency across all active courses.

3. Vidumurai – AI-Powered Leave Management System

React.js, Node.js, Firebase, AI Automation, REST APIs

- Developed a full-stack AI-powered web application automating student leave application and approval workflows, managing 500+ student records with real-time attendance validation logic.
- Implemented dual role-based dashboards (Student/Mentor) with automated notification triggers, reducing manual approval turnaround by ~60% and fully eliminating paper-based request handling.

4. Tourist Planner – ML-Powered Travel Recommendation System

Python, Streamlit, Scikit-learn, Pandas, Geospatial Libraries

- Built a content-based filtering recommendation engine with interactive geospatial map rendering, enabling discovery across 200+ destinations filtered by budget, region, and accessibility.
- Improved recommendation relevance by ~35% through feature engineering on user preference data and real-time filter optimization; deployed as a zero-setup Streamlit web app.

5. NeerNalan – Urban Water Governance System

Python, AI/ML, Geospatial APIs, WhatsApp Business API (SDG 6 Initiative)

- Developed a zero-hardware Digital Twin platform using AI and crowdsourced geospatial data, reducing municipal water crisis response times by **50%** across high-density urban wards.
- Engineered a triadic verification engine requiring three independent geospatial reports before triggering automated WhatsApp alerts, achieving **90% prediction accuracy** and eliminating false data submissions.

Work Experience

Ippopay Technologies Private LimitedDecember 2025 – February 2026

Frontend Development Intern — Fintech Production Environment

- Built and shipped 8+ responsive React UI components integrated with REST APIs for live fintech workflows, directly improving usability of customer-facing payment and onboarding modules.
- Optimized component rendering with lazy loading and code splitting, reducing average page load time by ~20% on key customer dashboards; maintained clean, modular architecture throughout.
- Participated in Agile sprints, daily standups, and code reviews within a team of 6 engineers, adhering to production-grade coding standards and Git branching workflows.

Lavo Technologies, TuticorinJuly 2024 – August 2024

Web Development Intern

- Designed and deployed a responsive multi-page B2B web application using JavaScript and REST APIs, digitizing 100+ monthly client enquiries through dynamic form integration and validation.
- Applied mobile-first, modular frontend architecture with cross-browser compatibility testing, delivering the full project on schedule within a 2-month engagement.

Achievements

- Secured the **IEEE EPICS Fund Grant of \$4,060** for the Hyaciguard project — awarded \$2,150 in initial funding with an additional \$1,910 grant based on demonstrated project impact and scalability.
- **Winner and finalist across multiple national-level hackathons and technical symposiums** — recognized for innovative engineering solutions, system-level thinking, and impactful project execution.

Education

Sri Sairam Engineering College, Chennai2022 – 2027 (Expected)

M.Tech – Computer Science and Engineering (Integrated) — CGPA: 7.9 / 10.0

- **Relevant Coursework:** Data Structures & Algorithms, Database Management Systems, Operating Systems, Computer Networks, Object-Oriented Programming, Software Engineering, Machine Learning

BMC Matriculation Higher Secondary School, Tuticorin

Higher Secondary Certificate (12th Grade) — Percentage: 80.3%